$= (A+B'+C+D)\cdot(A+B'+C+D')\cdot(A+B'+C)\cdot(A+B'+C')\cdot((A+C'+D')+B)\cdot((A+C'+D')+B')$

$\cdot(A'+B+C+D')\cdot((B+C'+D')+A)\cdot((B+C'+D')+A')$ (Postulate 4(b))

$=(A+B'+C+D)\cdot(A+B'+C+D')\cdot((A+B'+C)+0)\cdot((A+B'+C')+0)\cdot(B+(A+C'+D'))\cdot$

$(B'+(A+C'+D'))\cdot(A'+B+C+D')\cdot(A+(B+C'+D'))\cdot(A'+(B+C'+D'))$ (Postulate 2(a) and 3(a))

$=(A+B'+C+D)\cdot(A+B'+C+D')\cdot((A+B'+C)+D\cdot D')\cdot((A+B'+C')+D\cdot D')\cdot(B+(A+C'+D'))$

$\cdot(B'+(A+C'+D'))\cdot(A'+B+C+D')\cdot(A+(B+C'+D'))\cdot(A'+(B+C'+D'))$ (Postulate 5(b))

$=(A+B'+C+D)\cdot(A+B'+C+D')\cdot(A+B'+C+D)\cdot(A+B'+C+D')\cdot(A+B'+C'+D)\cdot(A+B'+C'+D')\cdot$

$(B+(A+C'+D'))\cdot(B'+(A+C'+D'))\cdot(A'+B+C+D')\cdot(A+(B+C'+D'))\cdot(A'+(B+C'+D'))$

(Postulate 4(b))

$=(A+B'+C+D)\cdot(A+B'+C+D')\cdot(A+B'+C+D)\cdot(A+B'+C+D')\cdot(A+B'+C'+D)\cdot(A+B'+C'+D')\cdot$

$((B+A)+(C'+D'))\cdot(B'+A+(C'+D'))\cdot(A'+B+C+D')\cdot(A+(B+C'+D'))\cdot(A'+(B+C'+D'))$

(Theorem 4(a))

$=(A+B'+C+D)\cdot(A+B'+C+D')\cdot(A+B'+C+D)\cdot(A+B'+C+D')\cdot(A+B'+C'+D)\cdot(A+B'+C'+D')\cdot(A+(B+C'+D'))\cdot(A'+(B+C'+D'))$

$(A+B)+(C'+D'))\cdot((A+B')+(C'+D'))\cdot(A'+B+C+D')$

(Postulate 3(a))

Removing common maxterms and placing them in order, we get:

$(A+B'+C+D)\cdot(A+B'+C+D')\cdot(A+B'+C'+D)\cdot(A+B+C+D')\cdot(A+B'+C+D)$

$(A'+B+C'+D')\cdot(A'+B+C+D')\cdot(A+B'+C'+D')\cdot(A+B'+C'+D)\cdot$ $\Pi(3,4,5,6,7,9,11)$

$\cdot(A+B+C'+D')\cdot$ $\Sigma(0,1,2,8,10,12,13,14,15)$

Sum of minterms: $\Sigma$

d) $F(A,B,C)=(A'+B)\cdot(B'+C)$

$=((A'+B)+0)\cdot((B'+C)+0)$ (Postulate 2(a))

$=((A'+B)+C\cdot C')\cdot((B'+C)+A\cdot A')$ (Postulate 5(b))

$=(A'+B+C)\cdot(A'+B+C')\cdot((B'+C)+A)\cdot((B'+C)+A')$ (Postulate 4(b))

$=(A'+B+C)\cdot(A'+B+C')\cdot(A+(B'+C))\cdot(A'+(B'+C))$ (Postulate 3(a))

$=(A'+B+C)\cdot(A'+B+C')\cdot(A+B'+C)\cdot(A'+B'+C)$ (Theorem 4(a))

Product of maxterms: $\Pi(4,5,2,6)$

Sum of minterms: $\Sigma(0,1,3,7)$